

Introduction to Research Methods

Science

May, 2026

Introduction to Research Methods (A37081)

Short Title: Introd to Research Methods
Faculty Science and Computing
Department Science
Cluster Placement and Projects
Author
Credits 5

Level: Intermediate

Description of Module / Aims

This module will introduce the student to techniques that are necessary in a research environment, such as accessing, referencing and utilising public literature and library resources. In addition, the student will be exposed to the necessary skills needed to compile a report to a scientific standard in order to prepare a literature review on a research topic.

Programmes

RESA-0042	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	3 / 6 / M
RESA-0042	BSc in Agricultural Science (ND) (WD_SAGRI_D)	3 / 5 / M
RESA-0042	BSc in Food Science with Business (WD_SFDSC_D)	3 / 6 / M
RESA-0042	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 5 / M

Indicative Content

- The scientific method, how to read and understand a scientific research article
- Search strategies and tools for searching the scientific literature, using online databases, books and Internet resources
- Referencing the literature using the Harvard and Vancouver systems & dedicated reference manager software.
- Identifying, evaluating and avoiding plagiarism in scientific writing
- Formatting and design of a literature review document using the built-in tools of MS Word
- Formatting tables, graphs, schemes and images appropriately for a research document
- Scientific writing skills
- Oral presentation of literature review

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use library resources to search the scientific literature and databases relevant to a chosen research topic.
2. Compile a review of the scientific literature on a specialised research topic in the format of a research report.
3. Apply the built-in tools of MS Word to compile, report and present a literature review on a specialised research topic.
4. Utilise reference manager software to support a literature review.
5. Communicate the research findings in an oral format.

Learning and Teaching Methods

- IT tutorials.
- Library training.

- Consultations with project supervisor.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Project</i>	70%	2,3,4
<i>Tutorial/Problem Sheets</i>	15%	1
<i>Presentation</i>	15%	5

Assessment Criteria

- <40%:** Inadequate understanding of the fundamentals of research methods and insufficient competence in their use. Insufficient level of knowledge of chosen research topic.
- 40%-49%:** Ability to show a basic knowledge of research methods and basic competency in their use. Basic level of knowledge of chosen research topic.
- 50%-59%:** As well as the above, the student will be able to demonstrate an understanding of some of the complexities of research methods and a good competency in their use. Display an adequate level of knowledge of chosen research topic supplemented by an ability to present this knowledge in a clear and concise manner.
- 60%-69%:** In addition to the above, the student will demonstrate a more detailed knowledge of research methods and a high level of competence in their use. Display a high level of knowledge of chosen research topic, supplemented by an ability to present this knowledge to a high technical standard.
- 70%-100%:** All of the above to an excellent level. In addition, the student will demonstrate a thorough level of knowledge of chosen research topic, supplemented by an ability to critically evaluate and discuss the topic in relation to a wider scientific context. The above advanced knowledge of the topic will be supported by superior reporting and presentation skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial	24	
Other	12	
Independent Learning	99	

Essential Material

- "Introduction to Research Methods Moodle page." <http://moodle.wit.ie>
- "SETU Waterford Library Catalogue." <https://librarywaterford.setu.ie/>
- "SETU Waterford Library Databases." <https://librarywaterford.setu.ie/databases/>

Supplementary Material

- Lebrun, J.L. *_Scientific writing 2.0: a reader's and writer's guide_*. Hackensack, N.J.: New World Scientific, 2011.
- Marder, M.P. *_Research methods for science_*. Cambridge: Cambridge University Press, 2011.

- Neville, C. _Complete guide to referencing and avoiding plagiarism_. Maidenhead , NY: Open University, 2010.

Requested Resources

- Room Type: Computer Lab